

This document defines the request and response contracts for the two endpoints that feed ticket creation (task 17) and reservation scheduling (task 20). It serves as the input specifications for backend information.

POST /v1/tickets

Description: Creates a new parking ticket when a vehicle enters the garage

Request Body:

```
{  
  "licensePlate": "ABC123",  
  "driverClass": "standard" | "accessibility" | "employee" | "eco",  
  "phone": "555-0199"           # optional - walk-up customers may omit.  
}
```

Success Response (201 Created):

```
{  
  "ticketId": "T-0042",  
  "licensePlate": "ABC123",  
  "phone": "555-0199",  
  "assignedFloor": 2,  
  "entryTime": "2026-03-11T09:15:00Z",  
  "status": "active"  
}
```

Request Field Details

Field	Type	Required	Notes
licensePlate	String	Yes	E.g. "ABC123"
driverClass	Enum	Yes	Standard, accessibility, employee, or eco status

phone	String	No	In person customers may omit
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Response Field Details

Field	Type	Required	Notes
ticketId	String	Yes	System-generated, e.g. "T-0042"
licensePlate	String	Yes	Echoed back from the request
phone	String	No	Only present if the customer provided it
assignedFloor	Int	Yes	E.g. 2, chosen by assignment algorithm
entryTime	String	Yes	ISO 8601 for standardized time record, server-generated timestamp of when the ticket was created
status	String	Yes	Always "active" on creation, it only changed to "closed" when the car exits via PUT /v1/tickets{id}/exit

Ticketing Error Responses

Code	Error Key	Trigger
400	missing_required_field	licensePlate, driverClass, or both, are absent fields
409	garage_full	No spaces available on any floor for the given driverClass
409	duplicate_plate	A ticket for that licensePlate already exists
500	server_error	Unexpected failure

POST /v1/reservations

Description: Creates a new reservation when an employee receives one

Request Body:

```
{
  "driverClass": "standard" | "accessibility" | "employee" | "eco",
  "phone": "555-0199",
  "scheduledArrival": "2026-12-03T10:00:00Z"
}
```

Success Response (201 Created):

```
{
  "reservationId": "R-0067",
  "assignedFloor": -1,
  "scheduledArrival": "2027-12-02T02:17:41Z",
  "status": "confirmed"
}
```

- Note that assignedFloor = -1. This is to indicate basement floor 1. With a common scale possibly looking like: -1, 1, 2. One basement floor, the ground floor (1), and the second floor.

Request Field Details

Field	Type	Required	Notes
driverClass	Enum	No	Standard, accessibility, employee, or eco status
scheduledArrival	String	Yes	ISO 8601 for standardized time record, an employee inputs the timestamp of when the reservation was scheduled for
phone	String	Yes	For SMS and lookup

Response Field Details

Field	Type	Required	Notes
reservationId	String	Yes	System-generated, e.g. "R-0021"
assignedFloor	Int	Yes	E.g. 2, chosen by assignment algorithm
scheduledArrival	String	Yes	Echoed back for visual confirmation
status	String	Yes	Always "confirmed" on creation, it only changed to "complete" when the car arrives via PUT /v1/reservations/{id}/check

Reservation Error Responses

Code	Error Key	Trigger
400	missing_required_field	phone, scheduledArrival, or both are absent fields
400	invalid_scheduled_arrival	Operator inputs a time from the past, or one too distant away in time
409	garage_full	No spaces available on any floor for the given driverClass
500	server_error	Unexpected failure